

# Low-risk upgrades for your Polymarket + Manifold prediction-market bot

## Context and quick diagnosis

Your 72-trade sample shows a “thin-edge + fat-tail” profile: modest positive average P&L (+\$0.24/trade) combined with rare, oversized losses (two positions totaling about -\$90) that dominate variance and likely crush risk-adjusted performance. Because the Sharpe ratio (risk-adjusted return) scales returns by volatility/standard deviation, eliminating “tail losses” can improve Sharpe far more than small tweaks to entry thresholds. <sup>1</sup>

Two implications follow from the mechanics of binary outcome tokens: if you’re long a side (YES or NO), the losing token can be worth \$0 at resolution (i.e., -100% on that leg), so **stop discipline must be automated**—not “remembered.” <sup>2</sup>

Separately, you’re currently trading as if each position is independent (fixed \$50 size, no portfolio rules). That is exactly how you end up with “surprise” correlated drawdowns (e.g., 10 markets that all hinge on the same macro move).

## Platform primitives that shape low-risk opportunities

### Polymarket: what enables true arbitrage (and what creates hidden risk)

Polymarket markets are binary with YES/NO outcomes, and markets are tradable via the CLOB when `enableOrderBook` is true; market metadata includes token IDs ( `clobTokenIds` ), liquidity and volume fields ( `liquidityNum` , `volume24hr` , etc.), plus fee flags ( `feesEnabled` ). <sup>3</sup>

The key arbitrage-enabling primitive is the **Conditional Token Framework** operations:

- **Split:** \$1 USDC.e → 1 YES + 1 NO token.
- **Merge:** 1 YES + 1 NO → \$1 USDC.e (requires equal quantities).
- **Redeem:** after resolution, winning tokens redeem for \$1 each. <sup>4</sup>

This makes “full-set” arbitrage possible: if you can acquire (YES + NO) for < \$1 total (net fees/slippage), you can often **merge immediately** to lock profit without waiting for resolution. <sup>5</sup>

Fees matter for arbitrage math. Polymarket states most markets are fee-free, but some market types have taker fees (notably crypto and certain sports), with `feesEnabled` indicating whether fees apply; fee schedules and formulas are documented. <sup>6</sup>

Resolution risk is the other “gotcha.” Polymarket documents a 2-hour challenge period after a proposal, and disputes can lead to longer timelines; the UMA adapter and UMA docs describe escalation to UMA’s DVM with 48–72 hour dispute resolution windows. <sup>7</sup> A widely reported 2025 incident illustrates that oracle/governance dynamics can produce controversial resolutions, which is especially relevant for “resolution arbitrage” strategies that rely on quick, unambiguous settlement. <sup>8</sup>

## Manifold: why “cross-platform arbitrage” is usually informational, not cash risk-free

Manifold’s currency **Mana** is explicitly described as play money that “cannot be converted to cash,” and Manifold’s terms emphasize it has no cash value and cannot be cashed out. <sup>9</sup> This means “arbitrage” between Manifold and Polymarket is generally **not** a guaranteed USD lock-in; it’s best treated as **cross-platform signal divergence** (informational arbitrage).

Manifold’s API explicitly permits “bots” and “automated trading systems” (within rate limits) and provides a `GET /v0/markets` endpoint with fields including `probability`, `totalLiquidity`, `volume`, and `volume24Hours`. <sup>10</sup>

## Ranked low-risk strategy upgrades you can ship this week

The ranking below prioritizes **(a)** Sharpe improvement per unit of engineering time, **(b)** low operational risk, and **(c)** alignment with 1–3 day holds. Sharpe deltas are estimates (you’ll firm them up by backtesting/paper trading), but they’re grounded in how Sharpe responds to variance reduction and to near-arbitrage payoff profiles. <sup>1</sup>

### Ranked strategies table

| Rank | Strategy (what you’ll code)                                                                  | Category  | Expected Sharpe improvement (estimate) | Implementation time | Risk level |
|------|----------------------------------------------------------------------------------------------|-----------|----------------------------------------|---------------------|------------|
| 1    | <b>Portfolio kill-switch:</b> trailing max drawdown + auto-liquidate + cooldown              | Risk      | <b>+0.8 to +2.0</b> (big variance cut) | 3–6 hours           | Low        |
| 2    | <b>Hard stop-loss enforcement:</b> bot-driven exits + “must-fill” logic                      | Risk      | <b>+0.6 to +1.8</b>                    | 0.5–1 day           | Low        |
| 3    | <b>Full-set CLOB arbitrage via merge:</b> detect $(askYES+askNO)<1$ , buy both, merge        | Arbitrage | <b>+0.7 to +1.6</b>                    | 1–2 days            | Low–Med    |
| 4    | <b>Fee-aware full-set arb in fee-enabled markets:</b> include taker fees, prefer maker fills | Arbitrage | <b>+0.4 to +1.2</b>                    | 1–2 days            | Med        |

| Rank | Strategy (what you'll code)                                                                                                          | Category          | Expected Sharpe improvement (estimate) | Implementation time | Risk level |
|------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------|---------------------|------------|
| 5    | <b>Combinatorial/linked-market arbitrage (heuristic):</b><br>dependent pairs/sets                                                    | Arbitrage         | <b>+0.4 to +1.3</b>                    | 2–4 days            | Med        |
| 6    | <b>Resolution-latency scanner:</b><br>“about to resolve” markets w/ objective sources                                                | Arbitrage         | <b>+0.3 to +1.0</b>                    | 1–3 days            | Med        |
| 7    | <b>Correlation caps:</b> cluster exposures, cap per “theme factor”                                                                   | Risk              | <b>+0.3 to +0.9</b>                    | 0.5–1 day           | Low        |
| 8    | <b>Volatility-adjusted sizing + stops</b> using <code>prices-history</code> realized vol                                             | Risk              | <b>+0.2 to +0.8</b>                    | 1 day               | Low        |
| 9    | <b>Liquidity + spread gating:</b><br>use <code>liquidityNum</code> , <code>volume24hr</code> , <code>spread</code> to filter entries | Entry             | <b>+0.2 to +0.6</b>                    | 3–6 hours           | Low        |
| 10   | <b>Cross-platform divergence (Manifold → Polymarket):</b><br>matching + discrepancy alerts                                           | Entry/<br>Arb-ish | <b>+0.1 to +0.8</b>                    | 1–2 days            | Med        |

## Strategy details (what to implement, why it's low-risk, pitfalls)

### 1) Portfolio kill-switch (trailing max drawdown + liquidate + cooldown)

Implement a portfolio layer that continuously tracks **equity peak** and **current drawdown** using realized + mark-to-market P&L. If drawdown exceeds X% (start conservative, like 3–5%), trigger: cancel open orders, market-out positions, and pause trading for N hours. This directly targets your “-90 disaster mode.” The reason this is high impact is mechanical: Sharpe improves if you reduce left-tail variance even if mean return stays similar. <sup>1</sup>

Implementation note: Polymarket supports bulk cancel endpoints (including cancel-all / cancel-market-orders) and market orders via “marketable limit orders,” enabling fast liquidation logic. <sup>11</sup>

### 2) Hard stop-loss enforcement (no discretion, no exceptions)

Your existing stop levels (-15/-25%) didn't hold because enforcement wasn't reliable. Replace “stop-loss as a rule” with “stop-loss as code”: each open position gets an exit plan that's executed automatically if P&L breaches limits. This is especially important because a losing outcome token can go to \$0 at resolution. <sup>2</sup>

Low-risk implementation pattern:

- Compute mark price via midpoint + spread (Polymarket exposes midpoint and spread endpoints). <sup>12</sup>
- If stop hit, place FOK/FAK orders to exit quickly, then “chase” once or twice if unfilled. Polymarket documents FOK/FAK as order types. <sup>13</sup>

### 3) Full-set CLOB arbitrage via merge (classic “YES+NO < \$1” scanner)

This is the cleanest “true arbitrage” you can do quickly on Polymarket:

- Read best asks for YES and NO for a condition.
- If `ask_yes + ask_no < 1 - buffer`, buy equal shares of both.
- Immediately call `mergePositions` to convert the complete set back to USDC.e, locking profit without waiting for resolution. Polymarket’s docs explicitly describe merge converting equal YES/NO into \$1 collateral. <sup>5</sup>

How to build it fast: use the official Python CLOB client (`py-clob-client`) for order placement and orderbook reads. <sup>14</sup>

Pitfalls: partial fills (only one side fills) turn “arbitrage” into a directional bet. Mitigation: submit both legs as FOK where possible, or set a timebox (e.g., 5–15 seconds) to cancel and revert if both don’t fill. Polymarket supports FOK/FAK semantics. <sup>13</sup>

### 4) Fee-aware full-set arb in fee-enabled markets (and optionally maker-biased execution)

In fee-enabled markets, the condition for “true arb” becomes:

`(cost_leg_A + cost_leg_B) + expected_fees + slippage < $1`.

Polymarket documents that some markets charge taker fees (notably crypto and some sports), provides the fee formula, and indicates `feesEnabled` on the market object. <sup>15</sup>

If you’re willing to place resting limit orders (maker) rather than crossing the spread (taker), you can reduce fee drag and sometimes qualify for maker rebates in eligible markets; the maker rebates program pays daily USDC rebates funded by taker fees. <sup>16</sup>

Risk is medium because the rebate economics add complexity and you can accumulate inventory unintentionally if one side fills.

### 5) Combinatorial/linked-market arbitrage (heuristics, not heavy ML)

A 2025 large-scale study of Polymarket data documented both intra-market (“market rebalancing”) and inter-market (“combinatorial”) arbitrage, estimating **\$40M** of extracted arbitrage profit over a one-year measurement window. <sup>17</sup>

You don’t need their LLM dependency detection to ship something this week. Start with 2–3 deterministic dependency templates:

- **Same event, alternate phrasing:** two markets that resolve simultaneously with logically complementary statements.
- **Implication pairs:** market A implies market B (e.g., “wins by 10+” implies “wins”).
- **Partition sets:** small sets that should sum to 1 within an event (when explicitly exhaustive in the market rules).

How to code:

- Use Polymarket’s Gamma API to group by event, end date, category, then screen candidate pairs. <sup>18</sup>
- Price candidates using the CLOB `book`, `midpoint`, `spread`, and `prices-history` endpoints. <sup>19</sup>

Risk is medium because mapping logic errors create unintended exposures.

### 6) Resolution-latency scanner (objective outcomes only, with dispute-aware timing)

The “resolution arb” that tends to be lowest-risk is not early/subjective politics—it’s **objective, already-known outcomes** (sports finals, published economic releases, etc.) where the market price lags public information. Low-risk implementation requires:

- Only consider markets whose resolution source is objective and public (read the rules). <sup>20</sup>

- Filter to markets near endDate, with large mispricing (e.g., YES at 0.90 when public data strongly implies near-certain), and sufficient liquidity. <sup>21</sup>

- Add a “dispute buffer”: Polymarket describes a 2-hour challenge period; disputes can extend timelines (48–72 hours via DVM). Your scanner should price capital lock time into expected return. <sup>7</sup>

Risk remains medium because of oracle/governance/high-ambiguity edge cases; a 2025 incident showed resolution controversy can happen, so restrict to the cleanest markets. <sup>22</sup>

## 7) Correlation caps (don't hold 10 “the same bet” at once)

Implement a simple exposure manager: define “themes” (BTC direction, a specific election, a single macro release) and cap total exposure per theme (e.g., max 20–30% of bankroll across all BTC-related markets). This is standard concentration risk control logic—basic but extremely effective at preventing correlated drawdowns. <sup>23</sup>

How to implement without ML:

- Use rules (tag/category, keyword matching on question text). Polymarket markets include category, tags, and question text. <sup>24</sup>

- Optionally add empirical correlation using recent price returns from `prices-history` (see strategy 8). <sup>25</sup>

## 8) Volatility-adjusted sizing + stops using realized vol

Percent stops (-15/-25%) ignore that different markets have different volatility regimes. Polymarket provides a historical price endpoint with selectable intervals, enabling a simple realized-vol estimate. <sup>25</sup>

A practical rule-set for binary prices (0–1):

- Compute  $\sigma$  of 1-minute returns over last 6–24 hours.

- Set stop distance = `k *  $\sigma$  * sqrt(H)` where H is your intended holding horizon in minutes/hours ( $k \approx 2$  is a conservative start).

- Position size = `risk_budget / stop_distance` with a hard cap.

This behaves like volatility targeting / volatility-based sizing: reduce size and tighten stops in high-vol markets. <sup>26</sup>

## 9) Liquidity + spread gating (stop trading bad microstructure)

This is low-hanging fruit that prevents both stop failures and slippage:

- Polymarket: require `liquidityNum` and `volume24hr` (or `volume1wk`) above thresholds; reject wide spreads using `spread` endpoint; ensure `acceptingOrders` and `enableOrderBook` are true. <sup>27</sup>

- Manifold: require `totalLiquidity`, `volume24Hours`, and/or `uniqueBettorCount` thresholds in addition to probability edge. <sup>28</sup>

This is low risk, fast to implement, and often improves realized performance purely by avoiding “untradeable” markets.

## 10) Cross-platform divergence scanner (Manifold → Polymarket), with webhook alerts

Because Mana cannot be converted to cash, cross-platform “arb” is primarily informational: use Manifold’s crowd forecast as an external signal to trade Polymarket when there’s a meaningful divergence, but only when Polymarket liquidity/spread are acceptable. <sup>29</sup>

Implementation outline:

- Pull Manifold markets (`GET /v0/markets`) and Polymarket markets (`GET /markets`, via Gamma). <sup>30</sup>

- Match by normalized question text + close/end date proximity ( $\pm 2$  days) + outcome-type compatibility.

- Compute divergence: `abs(P_poly_mid - P_manifold_prob)` and alert when it exceeds (say) 8–15 points **and** Polymarket market passes liquidity/spread filters. <sup>31</sup>

Alert delivery: send opportunities to Discord <sup>32</sup> or Slack <sup>33</sup> via incoming webhooks; or Telegram <sup>34</sup> via bot API. (Use alerts first; automate execution only after you see stable edge.)

## Implementation blueprint in Python

### Minimal architecture (fastest path to “ship this week”)

**Market ingestion layer** - Polymarket market discovery: Gamma `GET /events` / `GET /markets`; filter `active=true`, `closed=false`, and `enableOrderBook=true`. <sup>18</sup>

- Polymarket pricing: CLOB `GET /book`, `GET /midpoint`, `GET /spread`, `GET /prices-history`. <sup>35</sup>

- Manifold discovery: `GET /v0/markets` and extract `probability`, `totalLiquidity`, `volume24Hours`. <sup>36</sup>

**Realtime updates (optional but powerful)** - Use Polymarket’s market WebSocket: `wss://ws-subscriptions-clob.polymarket.com/ws/market` with asset IDs to receive orderbook/trade updates. <sup>37</sup>

**Opportunity engines (pluggable)** - Full-set arb engine (strategy 3/4)

- Linked-market arb engine (strategy 5)
- Resolution-latency engine (strategy 6)
- Cross-platform divergence engine (strategy 10)

**Execution layer** - Use official `py-clob-client` for order placement, cancellation, and auth setup. <sup>14</sup>

- Use Polymarket’s documented order types (FAK/FOK) for “must fill or cancel” behavior. <sup>13</sup>
- For merge-based arb: call merge operations on the CTF contract (your bot can do this on-chain; Polymarket documents merge mechanics and prerequisites). <sup>38</sup>

**Risk engine (runs before and after every action)** - Portfolio kill switch (strategy 1)

- Position stop enforcement (strategy 2)
- Concentration caps (strategy 7)
- Volatility sizing (strategy 8)

### A concrete “full-set arbitrage via merge” detection formula

For a candidate market (condition): - Let `a_yes` = best ask price for YES token

- Let `a_no` = best ask price for NO token

- Let `fee_est` = expected total fees (0 for fee-free markets; fee curve where `feesEnabled=true`) <sup>6</sup>

- Let `slip_buf` = your conservative slippage buffer (e.g., 0.002–0.01)

Trigger if:

`a_yes + a_no + fee_est + slip_buf < 1.0`

Then buy equal token quantity `q` on each side, and merge `q` sets for `q` USDC.e. <sup>5</sup>

## Guardrails specific to resolution and oracle risk

For any strategy that depends on resolution timing (especially strategy 6), bake in: - **Challenge-period clocking**: Polymarket documents a 2-hour challenge period after proposal. <sup>39</sup>

- **Dispute fallback**: UMA DVM can take 48–72 hours. <sup>40</sup>

- **Market selection**: prefer objective resolution sources and avoid “interpretation markets” where disputes are more likely; a 2025 controversy shows why this matters. <sup>41</sup>

## Metrics, testing, and “no more -\$90” operating rules

### What to log (minimum viable analytics)

Track, per trade and per day: - Gross and net P&L, and whether the trade was arb vs directional. - Max adverse excursion (MAE) to see whether stops are well-sized. - Slippage and fill time (especially for two-leg arb). - Equity curve + drawdown (for kill switch triggers). - Simple realized Sharpe on daily returns (even if noisy early). <sup>1</sup>

### Conservative default parameters to start (tune with data)

- **Risk per trade**: 0.25%–0.75% of bankroll (not fixed \$50).
- **Hard max loss per trade**: 1.0% bankroll (enforced via sizing + stop).
- **Portfolio max drawdown kill switch**: 3% (session) and 6–8% (weekly), with 12–24h cooldown.
- **Liquidity filters**: start with Polymarket `liquidityNum` and `volume24hr` thresholds that ensure stops can actually fill; both are present in market objects. <sup>42</sup>
- **Arb edge threshold**: require at least 0.3%–0.8% theoretical edge net fees/slippage before placing two-leg trades.

### “This week” build order (lowest risk path)

- 1) Implement portfolio kill switch + hard stop enforcement first (strategies 1–2).
- 2) Add liquidity/spread gating (strategy 9).
- 3) Only then deploy full-set merge arbitrage (strategy 3), starting tiny and requiring both legs to fill.
- 4) Add linked-market heuristics (strategy 5) only after you’ve eliminated operational losses.
- 5) Add cross-platform divergence alerts (strategy 10) as a research/idea generator, not as a primary “arb” profit engine, because Manifold is play-money. <sup>9</sup>

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<sup>1</sup> <sup>34</sup> <https://www.schwab.com/learn/story/calculate-sharpe-ratio-to-gauge-risk>

<https://www.schwab.com/learn/story/calculate-sharpe-ratio-to-gauge-risk>

<sup>2</sup> <sup>33</sup> Positions & Tokens

[https://docs.polymarket.com/concepts/positions-tokens?utm\\_source=chatgpt.com](https://docs.polymarket.com/concepts/positions-tokens?utm_source=chatgpt.com)

<sup>3</sup> <https://docs.polymarket.com/concepts/markets-events>

<https://docs.polymarket.com/concepts/markets-events>

<sup>4</sup> <https://docs.polymarket.com/concepts/positions-tokens>

<https://docs.polymarket.com/concepts/positions-tokens>

5 38 <https://docs.polymarket.com/trading/ctf/merge>

<https://docs.polymarket.com/trading/ctf/merge>

6 15 <https://docs.polymarket.com/trading/fees>

<https://docs.polymarket.com/trading/fees>

7 39 Resolution

[https://docs.polymarket.com/concepts/resolution?utm\\_source=chatgpt.com](https://docs.polymarket.com/concepts/resolution?utm_source=chatgpt.com)

8 22 32 41 <https://thedefiant.io/news/defi/polymarket-s-usd7m-ukraine-mineral-deal-debacle-traced-to-oracle-whale>

<https://thedefiant.io/news/defi/polymarket-s-usd7m-ukraine-mineral-deal-debacle-traced-to-oracle-whale>

9 29 Manifold FAQ

[https://docs.manifold.markets/faq?utm\\_source=chatgpt.com](https://docs.manifold.markets/faq?utm_source=chatgpt.com)

10 28 30 36 <https://docs.manifold.markets/api>

<https://docs.manifold.markets/api>

11 13 <https://docs.polymarket.com/trading/orders/create>

<https://docs.polymarket.com/trading/orders/create>

12 18 19 35 <https://docs.polymarket.com/market-data/overview>

<https://docs.polymarket.com/market-data/overview>

14 <https://github.com/Polymarket/py-clob-client>

<https://github.com/Polymarket/py-clob-client>

16 <https://docs.polymarket.com/market-makers/maker-rebates>

<https://docs.polymarket.com/market-makers/maker-rebates>

17 <https://suarez-tangil.networks.imdea.org/papers/2025aft-arbitrage.pdf>

<https://suarez-tangil.networks.imdea.org/papers/2025aft-arbitrage.pdf>

20 21 24 27 42 <https://docs.polymarket.com/api-reference/markets/get-market-by-id>

<https://docs.polymarket.com/api-reference/markets/get-market-by-id>

23 <https://www.bis.org/publ/bcbs63.pdf>

<https://www.bis.org/publ/bcbs63.pdf>

25 <https://docs.polymarket.com/api-reference/markets/get-prices-history>

<https://docs.polymarket.com/api-reference/markets/get-prices-history>

26 <https://us.plus500.com/en/newsandmarketinsights/position-sizing-guide>

<https://us.plus500.com/en/newsandmarketinsights/position-sizing-guide>

31 <https://docs.polymarket.com/trading/orderbook>

<https://docs.polymarket.com/trading/orderbook>

37 <https://docs.polymarket.com/market-data/websocket/market-channel>

<https://docs.polymarket.com/market-data/websocket/market-channel>

40 Polymarket/uma-ctf-adapter

[https://github.com/Polymarket/uma-ctf-adapter?utm\\_source=chatgpt.com](https://github.com/Polymarket/uma-ctf-adapter?utm_source=chatgpt.com)